

DEEPAN MUTHUVAIRAVAN

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SUMMARY

Cloud & Platform Operations Engineer | Linux Infrastructure Specialist | Multi-Cloud Administration (AWS | Azure | GCP)

18+ years of experience managing and supporting large-scale Linux and cloud infrastructures across AWS, Azure, and GCP. Multi-cloud engineer with production-scale AWS and GCP experience. Skilled in platform operations, system reliability, patching, monitoring, and automation using Terraform, Ansible, and standard CI/CD tools.

Proven expertise in maintaining secure, compliant, and highly available cloud environments. Experienced in Kubernetes and OpenShift operations, system upgrades, identity management, and infrastructure troubleshooting. Passionate about improving operational efficiency and platform stability through best practices and proactive monitoring.

CORE COMPETENCIES

- **Cloud Platforms:** AWS, Azure, GCP
 - **Infrastructure Operations:** Monitoring, Patching, Backup, Incident Response
 - **Automation & IaC:** Ansible, Terraform, Bicep
 - **Linux Systems:** RHEL, Ubuntu, CentOS, Debian
 - **Containers & Orchestration:** Kubernetes (EKS, AKS, GKE), OpenShift, Helm
 - **Networking & Security:** TCP/IP, DNS, IAM, Firewall, Load Balancing
 - **CI/CD & DevOps Tools:** Jenkins, GitHub Actions, Azure DevOps, GitLab
 - **Monitoring Tools:** CloudWatch, Azure Monitor, Prometheus, Grafana
 - **Process & Collaboration:** Agile, ITIL, ServiceNow, Jira, Confluence
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PROFESSIONAL EXPERIENCE

Freelance Linux Cloud/Platform engineer (Contract) | Independent Projects

Dec 2024 – Present | Scotland, UK (Remote)

- Delivered end-to-end Linux and cloud infrastructure support across AWS, Azure, and GCP.
- Managed hybrid Kubernetes workloads across EKS, AKS, and OpenShift environments.
- Automated infrastructure provisioning and configuration using Terraform and Ansible.
- Monitored and maintained uptime, resource usage, and security compliance across all environments.
- Integrated CI/CD systems for deployment automation and improved release stability.
- Supported DevOps tools such as Jenkins, GitHub Actions, and Azure DevOps.

Senior Linux Platform Engineer | Haveysoft

June 2023 – Sept 2024 | India (Remote)

- Performed Linux system administration, patching, and upgrade operations across hybrid environments.
- Automated standard infrastructure tasks with Ansible Tower to improve efficiency.
- Managed application deployments across AWS and Azure using Terraform and Helm.
- Ensured platform reliability, performance tuning, and incident management response.
- Collaborated with developers to maintain consistent environment configurations without direct coding.
- Implemented **KubeVirt-based virtualization on Kubernetes** to run full Linux virtual machines alongside container workloads, enabling hybrid VM + cloud-native architectures.
- Designed and managed **VirtualMachine (VM) and VirtualMachineInstance (VMI)** resources, including lifecycle operations (create, start, stop, restart) using Kubernetes-native workflows.
- Demonstrated **in-depth understanding of KubeVirt architecture**, including virt-operator, virt-controller, virt-handler, and virt-launcher components.
- Configured and validated **hardware-assisted virtualization (KVM)** on Kubernetes worker nodes, ensuring /dev/kvm availability and CPU virtualization support.

- Troubleshoot **KubeVirt console, VNC, and SSH access issues**, resolving RBAC, namespace scoping, and subresource permission challenges.

Senior Linux Cloud DevOps | Expedia Group UK

Feb 2022 – May 2023 | Lancesoft Inc India (Remote)

- Worked as part of client-facing DevOps and platform teams supporting cloud and container-based environments.
- Supported CI/CD and runtime compute platforms to ensure 99.9% service availability.
- Enhanced automation for infrastructure provisioning and compliance using Terraform and Bicep.
- Provided end-to-end operational support for Kubernetes and cloud-based workloads.
- Conducted incident management, RCA, and proactive monitoring using standard observability tools.
- Documented runbooks and best practices to improve developer onboarding and platform transparency.

Senior Software Engineer | Atos:

Apr 2017 – Jan 2022 | India (Remote)

- Provided Linux infrastructure support within large-scale managed services environments for enterprise clients.
- Supported Linux workloads running on VMware-based platforms, ensuring availability and stability of production systems.
- Participated in incident response, problem management, and root cause analysis (RCA) activities.
- Worked in environments with strict SLAs, operational governance, and customer-facing service delivery requirements.
- Collaborated with cross-functional teams to resolve complex infrastructure and platform issues.
- Gained experience supporting heterogeneous environments across Linux, virtualisation, and enterprise tooling.

Senior System Engineer | CGI:

Jun 2015 – Apr 2017 | India (Remote)

- Supported large-scale enterprise Linux environments hosted on VMware vSphere / ESXi platforms in 24x7 production.
- Performed day-to-day operational support for Linux virtual machines, including troubleshooting, performance analysis, and issue resolution.
- Worked closely with VMware infrastructure, storage, and network teams to resolve incidents impacting critical business services.
- Supported system patching, maintenance, and upgrade activities following strict change management and ITIL processes.
- Assisted with capacity planning and availability improvements for virtualised infrastructure.

Built strong foundations in enterprise operations, monitoring, and incident management.

Linux System Administrator | IBM: Delivered enterprise-grade Linux/Unix support, cluster configuration, and system performance optimization.

CERTIFICATIONS

- Microsoft Certified: Azure Administrator Associate (2021)
- Red Hat Certified Specialist in Ansible Automation (2018)
- Red Hat Certified System Administrator (2010)
- Kubernetes Administration Training (Completed)
- Terraform Associate Training (Completed)
- Azure Bicep Advanced Training (Completed)
- EMC Information Storage Associate V2 (2013)

Recent Hands-On Cloud Security & Access Architecture (GCP)

- Designed and implemented secure private access patterns for Google Cloud workloads using Identity-Aware Proxy (IAP), WireGuard VPN, and Bastion Host architectures.
- Implemented IAP-based SSH access to private Compute Engine instances without public IP exposure, leveraging IAM-based authentication and Cloud Audit Logs.
- Built a site-to-cloud WireGuard VPN solution enabling encrypted private network access from local workstations into GCP VPCs, with custom IP routing and firewall controls.
- Designed and deployed an enterprise-grade access model combining WireGuard VPN, hardened Bastion Host, and Google Cloud OS Login for identity-based SSH access.
- Enforced bastion-only access to private instances using network tags, restrictive firewall rules, and private IP routing.
- Integrated Cloud NAT to enable outbound API access for private bastion hosts without exposing inbound public connectivity.
- Hardened Linux SSH access by disabling password authentication, enforcing key-based and OS Login authentication, and restricting root access.
- Validated secure access paths through layered testing (VPN on/off, bastion-only SSH, direct access denial) ensuring least-privilege network design.
- Documented end-to-end secure access architectures and operational runbooks suitable for enterprise and regulated environments.

Recent Hands-On Platform Engineering & Security Projects

FastAPI – Automation & Platform APIs

- Designed and implemented RESTful APIs using FastAPI to expose infrastructure and automation workflows in a controlled and secure manner.
- Built FastAPI-based services for internal tooling, including request validation, structured responses, and robust error handling.
- Implemented API authentication and authorization mechanisms suitable for platform and automation use cases.
- Integrated FastAPI services with backend automation tools and scripts to enable Infrastructure-as-API patterns.
- Deployed and tested FastAPI applications using Uvicorn with environment-based configuration and logging.
- Applied API design best practices including schema validation (Pydantic), OpenAPI documentation, and versioning.

HashiCorp Vault – Secrets Management & Security

- Installed, configured, and operated HashiCorp Vault in lab environments to understand secure secret management patterns.
- Performed Vault initialization, unsealing, and access control configuration using policies and token-based authentication.
- Explored Vault authentication methods and secret engines applicable to infrastructure and application workflows.
- Practiced secure storage, retrieval, and rotation of secrets to eliminate hard-coded credentials.
- Integrated Vault concepts with Linux systems and cloud workflows to understand real-world secret distribution models.
- Applied security best practices including least-privilege access, audit logging, and separation of duties.
- Documented Vault setup procedures, usage patterns, and operational considerations for production readiness.

Operating Model & Reliability Focus

- Experience operating platforms in 24x7 production environments with on-call and incident response responsibilities.
- Strong understanding of SLAs, SLOs, and service availability targets in enterprise environments.
- Hands-on experience supporting incident management, root cause analysis (RCA), and post-incident reviews.
- Familiar with change management, release coordination, and operational governance in regulated organisations.
- Focused on platform stability, security, and operational excellence rather than feature-only delivery.

Security & Governance Awareness

- Strong awareness of security-by-design principles across cloud and Linux platforms.
- Experience aligning infrastructure and access controls with least-privilege and defence-in-depth models.
- Familiar with audit, logging, and traceability requirements in enterprise and regulated environments.
- Worked with IAM, network segmentation, and secure access patterns to reduce overall attack surface.
- Security-conscious approach to automation, avoiding credential sprawl and hard-coded secrets.

Hybrid Infrastructure & Migration Awareness

- Experience operating traditional VMware-based virtualised environments alongside modern cloud and container platforms.
- Strong conceptual understanding of VMware-to-cloud and VMware-to-container modernisation and migration approaches.
- Familiar with workload assessment, phased migration planning, and re-platforming strategies in enterprise environments.
- Applied automation, standardisation, and documentation practices to reduce operational complexity during platform transitions.

Right to Work & Availability

Right to Work: UK

Location: Scotland | Open to Remote / Hybrid | UK-wide roles